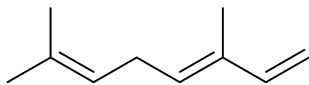


- 20** Ocimenes are a group of isomeric hydrocarbons with a sweet herbal scent and are commonly used in perfumes. The structure of one of its isomers is shown below.



β -ocimene

Which statements are correct?

- 1 β -ocimene has a total of four stereoisomers.
- 2 β -ocimene reacts with HBr to produce a major product containing two chiral centres.
- 3 β -ocimene undergoes both electrophilic addition and free radical substitution in the presence of excess bromine in the dark.
- 4 β -ocimene reacts with cold dilute alkaline potassium manganate(VII) to produce a major product with six -OH groups.

A 1 and 3

B 2 and 4

C 1, 2 and 3

D 1, 2, 3 and 4